



Society of Petroleum Engineers

SPE-226951-MS

Advancements in Downhole Monitoring for Horizontal Well Hydraulic Fracturing

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226951-MS](https://doi.org/10.2118/226951-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

As the primary downhole isolation tool enabling multi-stage hydraulic fracturing in horizontal shale gas wells, composite bridge plugs have become universally deployed throughout domestic shale gas operations. The dissolution kinetics and mechanical integrity of degradable bridge plugs exhibit intrinsic dependence on downhole environmental conditions, particularly formation temperature and annular pressure. Consequently, precise acquisition of real-time downhole temperature and pressure (T&P) profiles during plug deployment is essential for optimizing tool selection based on specific well conditions.

During high-rate fracturing operations, downhole pressure manifests dynamic fluctuations due to variable-rate fluid injection, proppant concentration changes, and near-wellbore friction effects. Simultaneously, temperature profiles are disrupted by the injection of large volumes of surface-temperature fracturing fluids into heated formations. Furthermore, reservoir heterogeneity induces complex fracture networks that create localized pressure sinks and thermal anomalies, generating significant deviations between actual downhole T&P conditions and predicted formation characteristics.

This study introduces an integrated monitoring solution featuring a dissolvable frac collet retrofitted with a intelligent frac ball which has high-resolution T&P sensing chip in it (design as [Fig. 1](#)). The corrosion-resistant sensor package, embedded within the collet assembly, continuously logs annular pressure (0-15,000 psi) and temperature (0-300°C) throughout the tool's operational lifecycle—from setting and fracturing through dissolution initiation. Engineered with high-stability MEMS piezoresistive pressure transducers and platinum RTD temperature sensors, the module captures data at programmable sampling intervals while withstanding extreme downhole vibrations.

Following fracture treatment completion, the controlled electrolytic dissolution of the magnesium-alloy collet and composite ball releases the encapsulated sensor package. This cylindrical module subsequently returns to surface entrained within flowback fluids, its buoyancy-enhanced design preventing settling in lateral sections. Surface recovery enables direct data extraction via USB interfaces, providing access to high-fidelity pressure transient records and thermal gradient profiles across all fracturing stages. Analysis of this dataset reveals critical behavioral patterns, including: Instantaneous shut-in pressure (ISIP) variations

between stages, Near-real-time fracture propagation signatures, Fluid thermal recovery profiles during soaking periods, Dissolution rate correlation with formation temperature.

The acquired downhole measurements enable quantitative validation of fracture models while informing future dissolvable tool selection based on actual well-specific conditions. This passive monitoring methodology eliminates wireline intervention risks and provides previously unattainable data from the fracture initiation zone throughout the entire plug functional cycle.

Introduction

During horizontal well volumetric fracturing operations, downhole pressure and temperature variations are critical to reservoir stimulation effectiveness, while their dynamic changes involve coordinated support from multiple raw materials, supplies, and downhole tools—including fracturing fluids, ceramic proppants, and dissolvable plugs—all significantly impacting stimulation outcomes. However, during current field operations, substantial fluid injection causes major deviations between actual downhole temperatures and formation temperatures. Simultaneously, heterogeneous fracture development creates dynamic discrepancies between downhole pressures and surface pump pressures. Existing technologies cannot effectively perform dynamic measurements of these parameters, resulting in non-targeted designs for fracturing fluids, proppants, and dissolvable tools, as well as lacking traceable methods for downhole tool selection or dynamic fracturing evaluation. Accurate temperature and pressure measurements hold vital significance: temperature dictates dissolution rates of fracturing materials (e.g., fluids, dissolvable plugs), enabling dissolution performance validation; pressure monitoring detects dissolution anomalies in all-metal dissolvable plugs and assesses pressure-bearing capacity, ensuring operational safety and equipment stability. These measurements provide critical data for process monitoring, control, and efficacy assessment while enabling fracturing design optimization, risk mitigation, and serving as critical references for tool/material R&D.

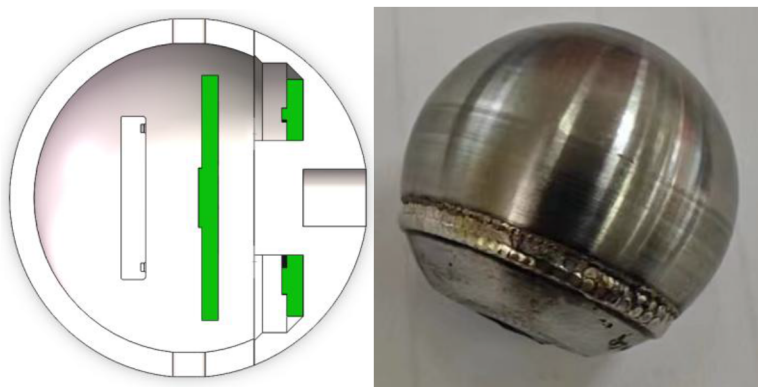


Figure 1—Internal structure diagram and overall appearance diagram of the intelligent fracturing ball

Current downhole temperature and pressure monitoring during fracturing operations primarily relies on fiber optic logging, wireline logging, and wireless transmission. For downhole pressure measurement in oilfields, conventional methods involve deploying pressure gauges to specific depths via wireline, drill pipe, or electric cables. These gauges are categorized as mechanical or electronic types. While mechanical gauges offer advantages such as ease of maintenance, high reliability, and shock resistance, they require retrieval after measurement periods, preventing real-time monitoring and necessitating well shutdowns that disrupt production. Electronic gauges feature extended service life and high precision, but suffer from susceptibility to downhole electromagnetic interference during data acquisition and transmission, leading to significant measurement errors.

Despite numerous temperature and pressure measurement devices being developed and deployed, existing technologies exhibit critical limitations: (1) Current single-point/multi-point sensors cannot achieve continuous distributed full-well monitoring, with additional measurement points increasing system complexity and costs; (2) Corrosive downhole fluids (particularly in CO₂ injection wells) damage traditional instruments, restricting them to short-term monitoring rather than permanent real-time deployment; (3) Conventional sensors cannot be permanently installed, requiring production interruptions that yield non-representative operational data (especially lacking for CO₂ injection temperature monitoring); (4) Electrically transmitted measurements remain vulnerable to electromagnetic interference.

Early downhole temperature measurement utilized thermocouples on production logging tools, providing only qualitative support for fluid profile analysis and cement bond evaluation, insufficient for quantitative interpretation. Permanent Downhole Gauges (PDGs) introduced in the 1960s faced high failure rates and situational inaccuracy. Quartz pressure sensors and Resistance Temperature Detectors (RTDs) adopted in the 1970s-80s improved PDG accuracy ($\pm 0.5^\circ\text{F}$ temperature precision) but were limited to single-point measurements below 140°C , unsuitable for deep or thermal recovery wells. Following the 1987 proposal of fiber optic temperature sensing, distributed temperature sensors were first installed at Shell Norway's Brunei field in 1993. Baker Hughes' mid-1990s "Intelligent Completion" concept accelerated temperature/pressure sensor deployment in offshore and high-yield onshore wells, with recent decades seeing integrated monitoring of pressure, temperature, flow rate, and multiphase flow.

Table 1—Statistics of Downhole Dynamic Monitoring Technologies

Measurement Instrument	Principle	Disadvantages
Fiber Optic Logging	The main principle is the temperature effect of optical fibers, which measures the temperature field in real time. FBG and EFPI fiber optic sensors are wavelength-encoded types, which are typically highly sensitive and can simultaneously measure parameters such as temperature and pressure.	High cost
Wireline Logging	Downhole parameter measurement via wireline operation	Long operation time and high cost
Wireless Transmission Monitoring Logging	Signals are transmitted between the surface and downhole, where the downhole controller collects parameters. The data is encoded into binary streams and transmitted according to relevant protocols.	Limited transmission range (within 3000 meters), unstable data transmission

Main technical challenges

Technical Challenges for Temperature-Pressure Sensing Chip

Temperature and pressure sensing chips, specifically designed for use with all-metal dissolvable plug technology, are mounted on the dissolvable plug and contained within the dissolvable ball.

As both the dissolvable ball and the dissolvable plug must possess sufficient pressure-bearing capacity during the fracturing process, and the sensing chip s must be capable of being flowed back to the wellhead during flowback, these sensing chip s are required to have a small size, high pressure resistance, and low density.

Consequently, the primary challenge in researching miniature temperature and pressure ball sensing chip s lies in designing small-sized sensing chip s capable of accurate temperature and pressure measurement under high-pressure and high-temperature (HPHT) downhole conditions.

Technical Challenges for Dissolvable Plug

Integrated with current field operational conditions, a plug tool with real-time temperature and pressure monitoring capabilities has been developed. This tool is designed to interface with existing plug-and-perf operations, providing critical technical support for fracturing design optimization, tool selection criteria, and ultimately enhancing the effectiveness of fracturing treatments.

Monitoring downhole temperature and pressure via this plug tool primarily presents the following three challenges:

Effective Integration of the Monitoring Sensing chip with Existing All-Metal Dissolvable plugs: How to achieve effective integration of the temperature and pressure monitoring sensing chip with the bridge plug through structural optimization and refinement of the existing all-metal dissolvable plug design.

Reliability of the Monitoring Sensing chip Under High-Temperature High-Pressure (HTHP) Downhole Conditions: How to ensure sustained reliability of the temperature and pressure monitoring sensing chip, maintaining its recording performance and data integrity under the high-temperature, high-pressure (HTHP) environment encountered during downhole operations.

To Surface and Data Readout: How to ensure the successful retrieval of the temperature and pressure monitoring sensing chip to the surface from the wellbore for data feedback.

To address these technical challenges, Vertechs introduced a new dissolvable collet has successfully developed an integrated temperature-pressure sensing chip for all-metal dissolvable collet, enabling full-lifecycle monitoring of downhole conditions. The intelligent frac ball represents a significant advancement in real-time downhole monitoring by eliminating the need for wireline tools. After dissolving post-frac, it flows back to the surface, where it can transmit pressure and temperature data.

Technical Approach for Temperature and Pressure Sensing Chip

The integrated hardware architecture of the temperature-pressure sensors chip requires high component density to ensure minimal footprint, primarily comprising a highly integrated System-on-Chip (SoC) and external functional circuits. The core processing unit employs a dsPIC33K SoC, supplemented by power management modules, temperature measurement circuits, communication interfaces, and battery power subsystems. This high integration enables compact form factor and minimal power consumption. As depicted in [Figure 3](#), the system block diagram encompasses power supply units, communication ports, operational amplifiers (op-amps), analog-to-digital converters (ADCs), FLASH memory storage, and temperature sensing components.

Low-energy Consumption Design

The core technical specifications of temperature and pressure sensors chip on low power consumption, with maximizing battery life through system-level software design being paramount. This study implements multi-mode operational states to differentiate power allocation, where granular state classification enables refined power distribution, minimizing energy loss.

The system operates through seven distinct modes:

1. Sleep mode halts the main controller while maintaining peripheral functionality, allowing interrupt or event-triggered wakeup.
2. Low-Power Run mode operates the core at minimum voltage, executes code from memory, restricts controller frequency, and reduces peripheral clock speeds.
3. Low-Power Sleep mode stops only the main controller clock, transitioning to Low-Power Run upon wakeup events.
4. Stop 0/1/2 modes achieve minimal power retention for memory and register contents by halting core-controlled clocks and high-frequency oscillators while maintaining low-frequency clock operation and real-time clock functionality, differing primarily in wakeup timing and conditions.

5. Standby mode employs internal controls for ultra-low power consumption, stopping all core-controlled clocks and oscillators while preserving internal clock operation; this mode clears volatile memory but retains backup register content, with wakeup possible via external reset, independent watchdog, wakeup pins, or internal clock events.
6. Shutdown mode achieves the lowest power state by terminating all core-controlled clocks and oscillators (both high/low frequency), erasing all non-backup memory and register data.
7. Run mode prioritizes maximum operational efficiency for the controller and peripherals at peak power consumption.

This system transitions among seven operational modes, with distinct tasks activating specific states: pressure acquisition operates in Run Mode; temperature collection engages Low-Power Run Mode; data storage utilizes Low-Power Run Mode; system standby enters Shutdown Mode; transitional periods employ Stop Mode for intermediate states.

An innovative design achieves an average operating current of approximately 30 μ A, ensuring prolonged functionality under limited power supply conditions.

Measurement Data Design

The microsphere-based monitoring system employs Microchip's dsPIC33CK Digital Signal Controller (DSC) as its core processor, specifically designed to meet stringent temperature and size constraints while delivering precision measurements.

Key Features of dsPIC33CK DSC:

1. Extended Thermal Tolerance
 - a. Operational range: -40°C to +150°C (validated per MIL-STD-810G)
 - b. Suitable for geothermal/high-PVT wells
2. High-Speed Data Acquisition
 - a. Triple 12-bit, 3.5 Msps ADC modules
 - b. Enables simultaneous multi-channel sampling
3. Signal Conditioning
 - a. Three independent operational amplifiers
 - b. On-chip programmable gain (1x-32x)
4. Data Storage
 - a. 256KB Flash memory for raw data logging
 - b. ECC protection against downhole radiation
5. Ultra-Low Power
 - a. 30 μ A standby current (Sleep mode)
 - b. Dynamic power scaling (1.8-3.6V) operation

The sputtered film core exhibits the following native characteristics:

Output sensitivity: 1.5 mV/V

Zero-point output: ± 10 mV

Output impedance: ≥ 1 k Ω

To amplify these weak signals for processing by the main controller, this study implements a three-op-amp instrumentation amplifier circuit utilizing the DSC's embedded operational amplifiers:

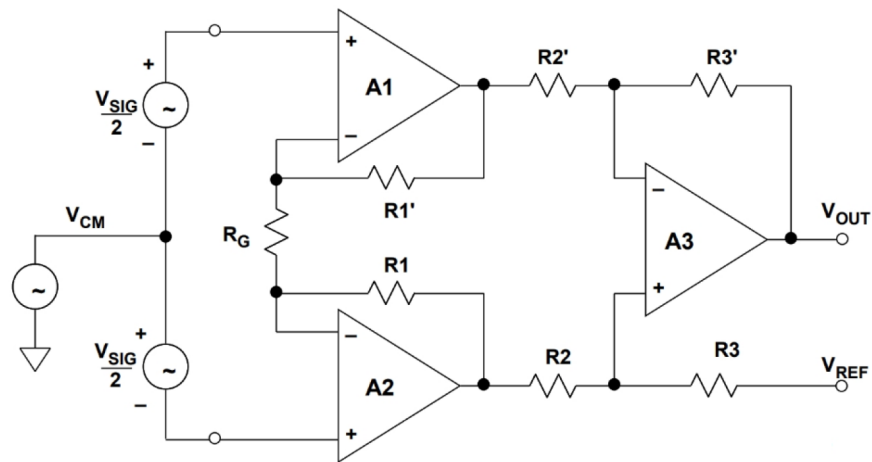


Figure 2—Signal conditioning circuit diagram

Simulated temperature and pressure test

Measuring the temperature sensor with a temperature box is an important step to ensure the accuracy and reliability of temperature measurement. In this paper, a temperature and pressure measuring ball with a battery was placed in a temperature chamber for simulation tests to verify the accuracy and reliability of temperature measurement. The following are the relevant scheme steps for the temperature chamber simulation test:

Indicators of test equipment. High and low temperature environmental simulation test chamber: It ensures that the required temperature range and accuracy requirements can be met, with temperature fluctuation $\leq \pm 0.5^{\circ}\text{C}$ and temperature uniformity $\leq 2^{\circ}\text{C}$. Temperature sensors and data acquisition systems: used for real-time monitoring and recording of temperature changes during the test process. Voltage stabilizing power supplies and other auxiliary equipment: Ensure the stability and safety of the power supply during the test process.

Incubator. The model of the incubator used in this experiment is: ETE-KWB=1000L. This incubator features a fast temperature change rate and good linearity. It is applied to the rapid temperature change experiments of electronic products, components and aerospace materials as well as profit screening tests.

Through long-term measurement of the stepwise temperature rise test of the temperature box, the target working stability of this temperature box was set at 115°C , and the measured stable temperature was 115°C . It can prove that the temperature measurement is accurate with an error of less than 0.5°C .

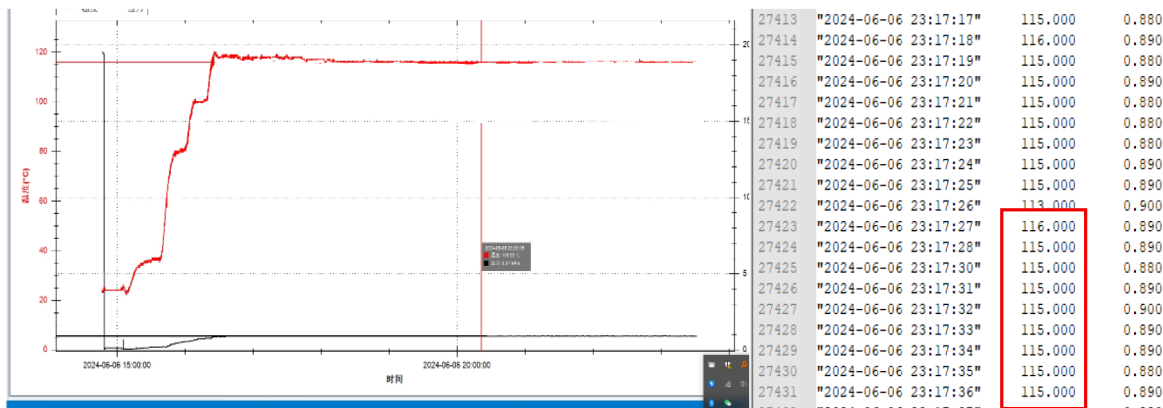


Figure 3—Temperature measurement curve and data of the temperature chamber

Simulated pressure test

A simulated pressure test was conducted in the simulated pressure test device. The steps for the simulated test pressure are as follows:

1. Put the temperature and pressure measuring balls into the casing.
2. Casing emptying;
3. The pressure increase value is 50MPa and remains stable for 10 minutes.
4. Increase the pressure to 100MPa and stabilize for 15 minutes.

Through pressure tests, it is proved that the measured pressure value differs from the applied pressure value by no more than 0.5MPa.



Figure 4—Simulate the pressure test curve of the pressure testing machine

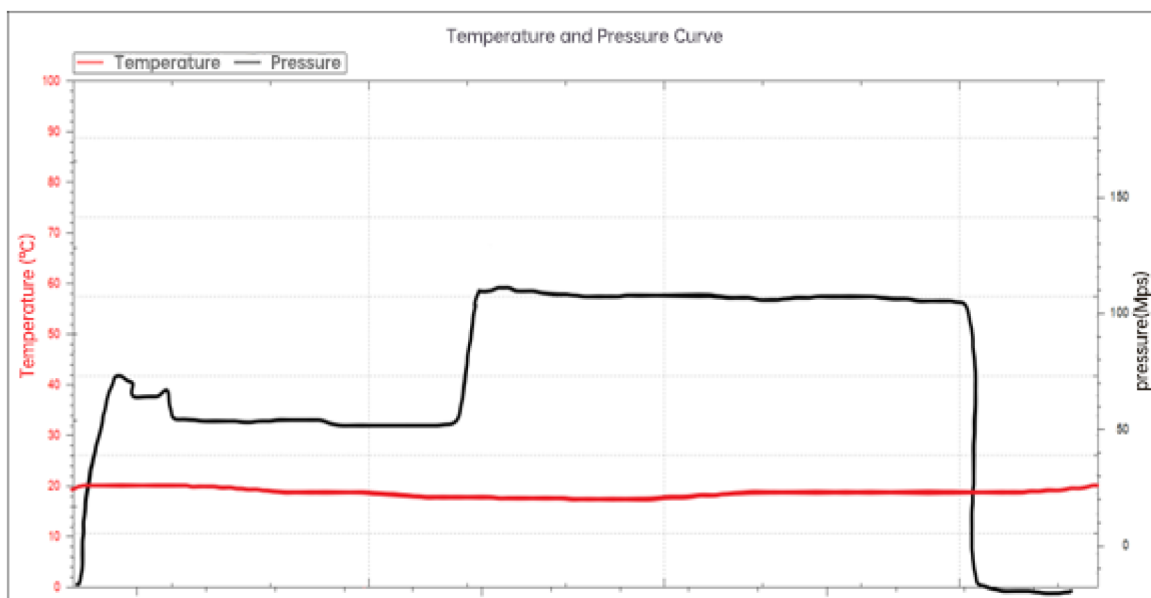


Figure 5—Pressure measurement curve and data of the temperature-pressure sensing chip

Storage Capacity Test

This test adopted a centralized data collection space with a collection frequency of 1 second. After collecting 28,210 data points through this test, the system entered a stop collection phase, proving that the system can store more than 28,000 points.

1	Time	Temperature(°C)	Pressure(Mpa)
2	"2024-06-06 14:46:46"	23.000	19.500
3	"2024-06-06 14:46:47"	23.000	19.520
4	"2024-06-06 14:46:48"	23.000	19.520
5	"2024-06-06 14:46:49"	24.000	19.510
6	"2024-06-06 14:46:50"	23.000	19.510
7	"2024-06-06 14:46:52"	23.000	19.520
8	"2024-06-06 14:46:53"	23.000	19.510
28191	"2024-06-06 23:31:21"	115.000	0.880
28192	"2024-06-06 23:31:22"	115.000	0.890
28193	"2024-06-06 23:31:23"	115.000	0.890
28194	"2024-06-06 23:31:24"	115.000	0.890
28195	"2024-06-06 23:31:25"	115.000	0.890
28196	"2024-06-06 23:31:26"	115.000	0.890
28197	"2024-06-06 23:31:28"	115.000	0.890
28198	"2024-06-06 23:31:29"	115.000	0.880
28199	"2024-06-06 23:31:30"	115.000	0.890
28200	"2024-06-06 23:31:31"	115.000	0.890
28201	"2024-06-06 23:31:32"	115.000	0.900
28202	"2024-06-06 23:31:33"	115.000	0.890
28203	"2024-06-06 23:31:34"	115.000	0.890
28204	"2024-06-06 23:31:35"	115.000	0.890
28205	"2024-06-06 23:31:36"	115.000	0.890
28206	"2024-06-06 23:31:37"	115.000	0.890
28207	"2024-06-06 23:31:38"	115.000	0.890
28208	"2024-06-06 23:31:39"	115.000	0.890
28209	"2024-06-06 23:31:41"	115.000	0.890
28210	"2024-06-06 23:31:42"	115.000	0.890
28211	"2024-06-06 23:31:43"	115.000	0.890

Figure 6—Storage capacity test results

Technical Approach for dissolvable collet

The dissolvable collet combines anchoring and sealing functions into a single component using high-expansion dissolvable metal to create a collet structure with slip segments and a metal seal ring, eliminating the need for elastomers as a sealing material.

Pressure Test

To validate the field applicability of dissolvable collet and temperature-pressure monitoring devices, extensive laboratory tests were conducted. The results demonstrate that the system meets operational requirements with 70 MPa pressure-bearing capacity and 150°C temperature resistance.



Figure 7—RIH Wireline tools schematic diagram

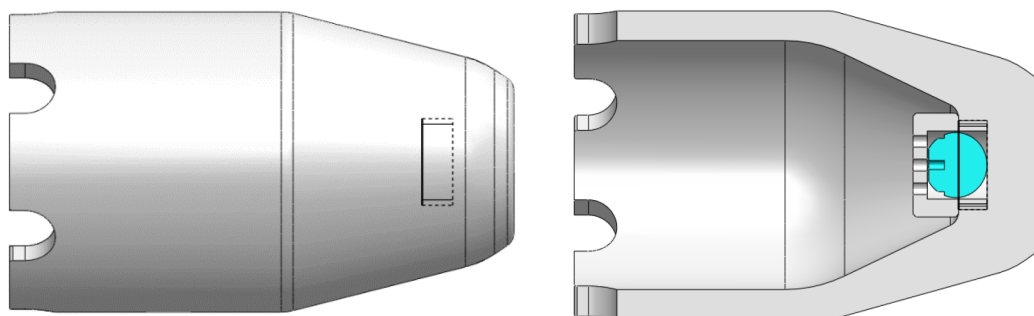


Figure 8—Intelligent frac ball and dissolvable dart

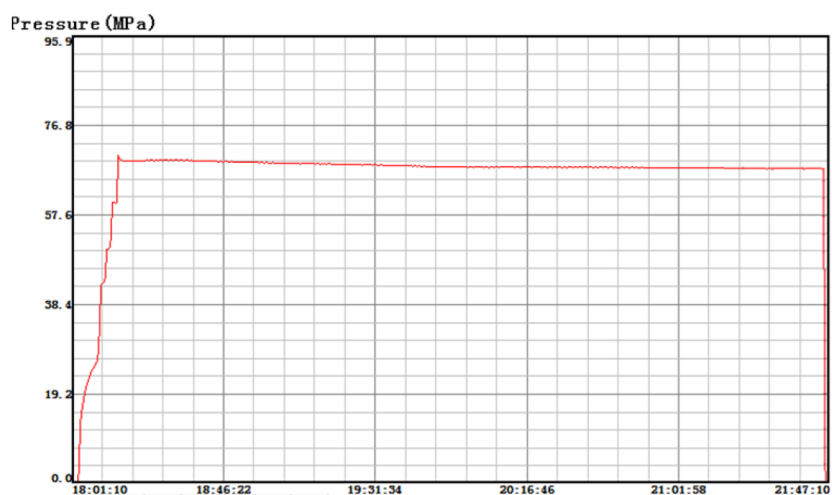


Figure 9—150°C, 10,000Psi pressure test

Frac Ball Packaging Performance Test

Currently, through the encapsulating mechanism, the frac ball are placed inside the encapsulating mechanism with appropriate sizing. Additionally, there are 7 openings on the top of the encapsulating mechanism, which allow the frac ball inside the encapsulating mechanism to effectively monitor pressure and temperature during the pressure-bearing process in the wellbore.

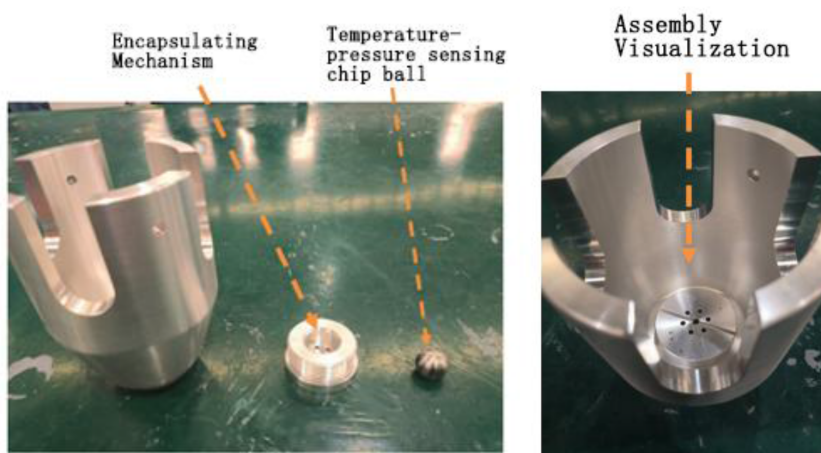


Figure 10—intelligent frac ball packaging performance test

Case Study

Field Test Conducted on June 21, 2024 at Well NT-1H (The well geological design indicates a static formation temperature of 105°C)

A field validation was performed by integrating the microsphere-based monitoring system within dissolvable collet tools to achieve real-time pressure and temperature measurements during fracturing operations. It shows that that pressure exceeded 120 MPa and fracturing fluid temperature peaked at 80°C during fracturing operations.

Comparative analysis of fracturing fluid temperatures acquired during operations versus static formation temperatures confirms that Continuous fluid injection dynamics induces significant thermal decay in fracturing fluids, with recorded temperatures substantially below virgin formation temperatures during active fracturing operations. This well exhibits a 23.8% temperature differential.



Figure 11—Field Test Field On-site image

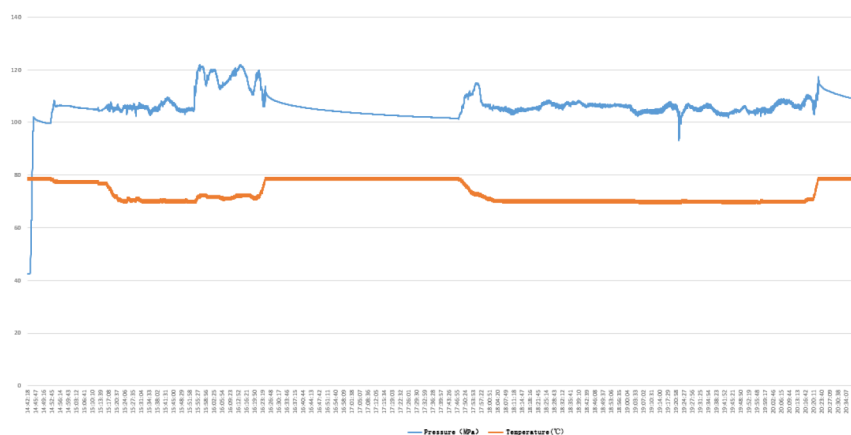


Figure 12—Field Test Results.

Conclusion

The intelligent frac ball represents a significant advancement in real-time downhole monitoring by eliminating the need for wireline tools. After dissolving post-frac, it flows back to the surface, where it can transmit pressure and temperature data. Equipped with operation configuration software, it enables functions such as parameter configuration, data reading, and analysis, providing a seamless, continuous monitoring solution for optimizing frac operations without the complexities of traditional methods.

Up to now, The dissolvable intelligent frac ball has been successfully deployed in hundreds of wells across China. Downhole monitoring confirms significant thermal decay in fracturing fluid during active operations due to continuous injection fluid dynamics, with recorded temperatures substantially lower than native formation temperatures. Comprehensive field data analysis reveals a critical pattern: sustained temperature drops of 70–85°F (21–29°C) are consistently observed when static bottomhole temperature (BHT) is

below 194°F (90°C), while the thermal differential widens to 100–120°F (38–49°C) when static BHT exceeds 248°F (120°C). Extreme-condition validation testing (pressure differential: 14,500 psi / 100 MPa, temperature: 260°F / 127°C) further substantiates its reliability. Advances in dissolvable plug technology have eliminated milling interventions. Post-fracturing plug dissolution enables smart unit flowback to surface. Operational configuration software supports parameter setup, data retrieval, and analysis. This cable-free methodology delivers continuous monitoring, offering a streamlined, comprehensive solution for fracturing optimization.

Acknowledgments

We wish to extend our sincere appreciation to Vertechs Group for its support and permission to publish this work.

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